

- 14** An ideal gas in a container of fixed volume 1.0 m^3 has a pressure of $3.0 \times 10^5 \text{ Pa}$ at a temperature of 200 K . The gas is heated until the temperature reaches 400 K . Some gas is released from the container during the heating to keep the pressure constant.

What volume does the gas released from the container occupy, if it is at atmospheric pressure of $1.0 \times 10^5 \text{ Pa}$ and at a room temperature of 300 K ?

- A** 0.500 m^3 **B** 2.00 m^3 **C** 2.25 m^3 **D** 4.50 m^3

Ans C

$$n_1 = \frac{p_1 V_1}{RT_1} = \frac{3.0 \times 10^5 (1)}{8.31(200)} = 180.5 \text{ mol}$$

$$n_2 = \frac{p_2 V_2}{RT_2} = \frac{3.0 \times 10^5 (1)}{8.31(400)} = 90.25 \text{ mol}$$

$$n_{\text{released}} = n_1 - n_2 = 90.25 \text{ mol}$$

$$V_{\text{released}} = \frac{n_{\text{released}} RT}{p} = \frac{90.25(8.31)(300)}{1.0 \times 10^5} = 2.25 \text{ m}^3$$

- 15** When a volatile liquid evaporates it cools down